1. Competency #5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- Highly relevant: This is the core skill tested.
- Students must identify the repeating annual dynamic mechanism:
  - All existing penguins (regardless of age) can reproduce (since "one-year-olds can breed" and there's no upper age limit or mortality-induced generational gap);
  - Each pair → 1 chick ⇒ if the current population is N (even), there are $N/2$ pairs, producing $N/2$ new chicks;
  - Total before mortality: $N + N/2 = 1.5N$;
    After 20% die, 80% survive: $0.8 \times 1.5N = 1.2N$.
- Thus, the population multiplies by a constant factor of 1.2 each year, forming a geometric sequence: $P = 10{,}000 \times (1.2)^7$.
- Although the options express this as $(1.5 \times 0.8)^7$ (which equals 1.2^7), the key insight is recognizing the invariant growth rate as a structural pattern.

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2. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying significant variables

> *"Identify and describe the mathematical aspects... including identifying the significant variables"*

- Highly relevant: Students must clearly define:
  - Initial value: 10,000;
  - State variable: total penguin count at the start of year t , denoted N_t ;
  - Rule-based parameters: reproduction rate (2 penguins → 1 chick ⇒ growth factor 1.5), survival rate (0.8);
  - Time horizon: 7 years.
- Crucially, they must understand that all penguins reproduce (due to early maturity and no generational loss), meaning growth depends on total population, not just adults—this is the decisive modeling insight.

3. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

> *"Simplify or decompose a situation... to make it amenable to mathematical analysis"*

- Strongly relevant: Students must reduce the complex life cycle to an annual update rule:
  1. Start of year: N penguins;
  2. Spring: add $N/2$ chicks → total = $1.5N$;
  3. End of year: 80% survive → $N_{\text{next}} = 1.2N$.
- This discrete-time, annual-step abstraction ignores real-world complexities like sex ratio fluctuations or stochastic mortality—exactly the kind of simplification needed for tractable modeling.

4. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- Strongly relevant: The stem explicitly lists four assumptions. Students must recognize how they jointly enable the simple model:
  - "One-year-olds can reproduce" ⇒ no generational delay;
  - "20% mortality applies to all individuals" ⇒ no age-specific death rates;
  - "Equal sex ratio each year" ⇒ maximum pairing efficiency.
- Together, these justify the constant per-capita growth factor. If any assumption were violated (e.g., chicks couldn't breed in their first year), the model would be fundamentally different.

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5. Competency #9: Translate a problem into a standard mathematical representation or algorithm

> *"Translate a problem into a standard mathematical representation or algorithm"*

- Moderately relevant: Students must encode the mechanism as an exponential function:
  $P = P_0 \times r^t$, where $r = 1.2$ and $t = 7$.
- Although the options present it as a product (e.g., 1.5×0.8), the essence is translating verbal rules into algebraic form, a standard modeling output.

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6. Competency #1: Select a mathematical description or representation that describes a problem

> *"Select a mathematical description or a representation that describes a problem"*

- Moderately relevant: Students must judge that this scenario is best modeled by a recursive relation or exponential function, not differential equations (continuous time) or manual year-by-year tables (inefficient).
- Identifying the correct structure among given options also involves evaluating and selecting an appropriate representation.

1. 第5项：识别问题或情境中的数学结构（包括规律、关系和模式）

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- 高度相关：这是本题的核心能力。
- 学生需从文字描述中识别出年度重复的动态机制：
  - 每年所有现存企鹅（无论年龄）都能繁殖（因“一岁即可繁殖”且无寿命上限）；
  - 每对 → 1 雏 ⇒ 若当前有 N 只企鹅（ N 为偶数），则有 $N/2$ 对，新增 $N/2$ 雏鸟；
  - 年末总数量变为 $N + N/2 = 1.5N$ ，再经 20% 死亡 → 存活 $0.8 \times 1.5N = 1.2N$ 。
- 因此，每年种群乘以固定因子 1.2，形成几何序列： $P = 10000 \times (1.2)^7$ 。
- 虽然选项写成 $(1.2 \times 0.8)^7$ （实际应为 1.2^7 ，但 D 选项数值上等价于 $(1.2)^7$ 因其隐含 $1.5 \times 0.8 = 1.2$ ），关键在于识别恒定增长率这一结构性规律。

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2. 第6项：识别并描述现实问题中的数学要素，包括关键变量

> *"Identify and describe the mathematical aspects... including identifying the significant variables"*

- 高度相关：学生必须明确：
  - 初始值：10,000；
  - 状态变量：每年初的总企鹅数 N_t ；
  - 规则变量：繁殖率（每2只产1雏 → 增长因子 1.5）、存活率（0.8）；
  - 时间变量：7 年。
- 尤其关键的是理解“所有企鹅都能繁殖”（因一岁即可育且无死亡导致断代），这决定了增长是基于总数量而非仅成年个体——这是建模成败的关键洞察。

3. 第7项：简化或分解问题，使其适合数学分析

> *"Simplify or decompose a situation... to make it amenable to mathematical analysis"*

- 强相关：学生需将复杂的生物生命周期简化为年度更新规则：
  1. 年初： N 只；
  2. 春天：新增 $N/2$ 雏 → 总数 $1.5N$ ；
  3. 年底：80% 存活 → $N_{\text{next}} = 1.2N$ 。
- 这种年度离散时间步长的抽象，忽略性别比例波动、随机死亡等现实复杂性，正是建模所需的简化。

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4. 第12项：识别模型中的约束、假设与简化

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- 强相关：题目明确列出四项假设，学生必须意识到：
  - “一岁即可繁殖”意味着无世代延迟；
  - “20% 死亡率适用于所有个体”意味着无年龄特异性死亡；
  - “每年初性别比1:1”确保最大繁殖效率。
- 这些假设共同支撑了“恒定增长因子”这一核心简化。若任一假设不成立（如雏鸟不能当年繁殖），模型将完全不同。

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5. 第9项：将问题转化为标准数学表示或算法

> *"Translate a problem into a standard mathematical representation or algorithm"*

- 中等相关：学生需将上述机制转化为指数函数形式：
  $P = P_0 \times r^t$ ，其中 $r = 1.2$ ， $t = 7$ 。
- 虽然选项以乘积形式呈现（如 1.5×0.8 ），但本质是将文字规则编码为代数表达式，属于标准建模输出。

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6. 第1项：选择能描述问题的数学表述或表示方式

> *"Select a mathematical description or a representation that describes a problem"*

- 中等相关：学生需判断：此问题适合用递推关系或指数模型，而非微分方程（连续时间）或表格枚举（低效）。
- 在给定选项中识别正确结构，也涉及表示形式的选择与验证。